

AI-NATIVE ERP

Master Product, Functional, Technical & Implementation Blueprint

RUN THE BUSINESS BY ASKING

DESIGN NORTH STAR	PRODUCT PROMISE
Easy enough for a first-time user	Every screen can explain itself
AI present in every module	Every answer can lead to a safe action
Operate in the user's mother tongue	Owners can ask for any report conversationally
Human control for business impact	Permissions, validation, approval and audit always apply

Prepared as the successor to the Complete ERP Blueprint. This document defines an AI-native operating system for multi-company businesses, not a conventional ERP with a chatbot added later.

Executive Blueprint

The product is an AI-native ERP in which navigation, learning, reporting, data entry, exception handling and business decisions can all begin with natural language. The conventional user interface remains available, but the primary interaction pattern is Ask - Understand - Prepare - Confirm - Act.

The platform must serve a first-time employee who needs step-by-step guidance, an experienced operator who wants speed, and an owner who wants answers without building reports.

Product position

An ERP anyone can operate by talking to it. The defensible advantage is not the language model alone; it is the combination of ERP-grade data, deterministic business rules, safe tools, multilingual interaction and complete auditability.

Non-negotiable outcomes

- A new user completes core tasks without classroom training.
- Every module exposes contextual help, analytics and safe AI actions.
- Users ask and receive answers in Malayalam, English, Hindi, Tamil, Arabic and additional configured languages, including mixed-language input.
- AI never bypasses permissions, validations, posting rules or approval workflows.
- Every material claim includes scope, filters, freshness and a drill-down path.
- Every AI-initiated transaction is traceable from request to result.

Primary personas

Persona	Need	AI-native experience
Owner / CEO	Know what needs attention	Daily brief, conversational KPI analysis, anomaly explanations
Admin	Configure the system	Natural-language setup, workflows, roles and dashboards
Manager	Decide and approve	Decision packs, risk flags, comparisons and approvals
Operator	Complete tasks quickly	Command bar, guided forms, error resolution and voice
Accountant	Trust the numbers	Reconciliation assistance, explainable variances and close checks
New employee	Learn while doing	Teach Me mode with screen-aware, role-aware instructions

1. Product Principles & Success Measures

Principle	Design rule	Measurable target
Zero-friction use	Progressive disclosure; ask only for required fields	80% of common tasks completed without training
AI everywhere	One consistent Ask ERP surface on every screen	100% of production modules publish help, read and draft capabilities
Mother-tongue operation	Text and voice preserve business meaning	Top launch languages pass intent and terminology evaluation
Safe action	AI calls governed domain APIs only	0 direct model-to-database writes
Evidence first	Reports show period, entity, currency and source	100% of numeric answers carry scope metadata
Human accountability	High-impact actions require explicit confirmation	100% L3 actions have confirmation and audit evidence
Graceful fallback	Uncertain AI asks or routes to conventional UI	No fabricated IDs, balances or transaction status

North-star metrics

- Time-to-first-value: company configured and first valid invoice created within one working day.
- Task success rate: at least 95% for the top 25 role-specific workflows.
- Median Ask ERP response: under 3 seconds for cached help and under 8 seconds for normal analytics.
- AI action correction rate: less than 2% after user confirmation for supported draft actions.
- Support deflection: at least 50% of how-to questions resolved inside the product.
- Trust: users can reproduce every reported number by opening the exact underlying report.

MVP definition

The first sellable release includes organization and access, CRM, sales, procurement, inventory, finance essentials, reporting, workflow, audit, multilingual Ask ERP, contextual help, conversational read queries and draft creation for the highest-volume transactions.

2. Experience Blueprint: Ask ERP

Ask ERP is a persistent command surface available from the global header, every module, record side panel and mobile application. It combines global search, navigation, training, reporting and actions.

Mode	User request	System behavior
Navigate	Open overdue invoices	Opens the correct filtered list
Learn	How do I make a stock transfer?	Explains and highlights the next valid control
Diagnose	Why can't I submit this PO?	Checks the record and lists missing or invalid fields
Report	This month's sales by branch	Runs a governed semantic query and shows totals plus drill-down
Prepare	Create a quotation for Rahman Traders	Builds a draft using customer pricing and live stock
Act	Submit this for approval	Shows impact, requests confirmation and invokes workflow
Automate	Every morning alert collectors about large overdue invoices	Builds a previewable trigger-condition-action rule

Screen context contract

Each AI request carries user, tenant, company, branch, role, locale, module, screen, record identifier, field focus, permitted actions, conversation state and a short-lived context version. Sensitive values are minimized before they reach the model.

Response contract

- Answer in the user's requested language and preserve domain terms where translation may create ambiguity.
- Display data scope: company, branch, period, currency, status and last refresh time.
- Offer no more than five relevant next actions.
- Mark estimates, forecasts and inferred explanations explicitly.
- For actions, show a structured preview of proposed changes before confirmation.
- If confidence is insufficient, ask a focused clarification or open the safest conventional workflow.

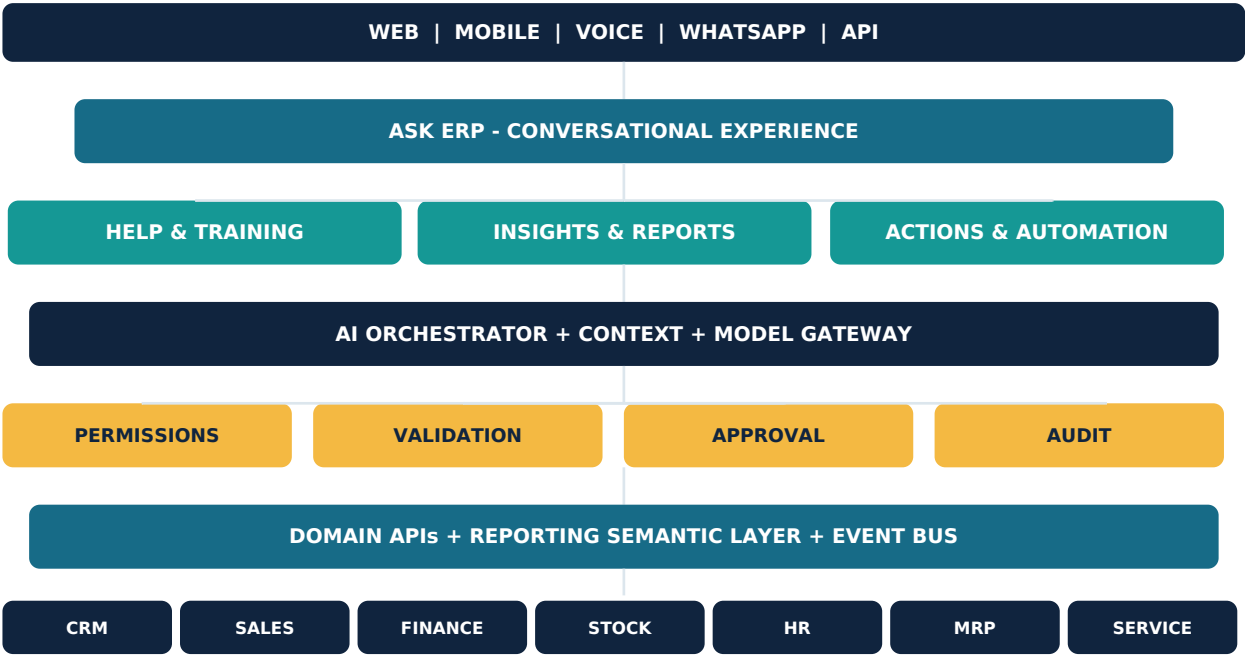
Teach Me mode

The assistant converts a business goal into a short guided sequence. Each step highlights one field or action, explains why it matters, validates completion and resumes from the last successful step. Instructions must reflect the current role, configuration and record—not a generic manual.

Smart forms

Common records can be created conversationally. AI gathers only required information, resolves master data candidates, applies defaults, validates tax and pricing, then presents the complete draft. Advanced fields remain accessible.

3. Reference Architecture



The core boundary is the Tool Gateway. Models can reason and propose, but only deterministic services can read governed data or change business state. Domain APIs remain the system of record.

Layer	Responsibilities
Experience	Web, mobile, voice, WhatsApp, global command bar, contextual side panel
AI gateway	Authentication, rate limits, model routing, prompt policy, streaming and cost controls
Orchestrator	Intent classification, plan construction, tool selection, retries and response assembly
Context engine	Screen, role, tenant, terminology, history, permissions and relevant knowledge
Knowledge / RAG	Product help, company SOPs, policies, chart of accounts guidance and resolved tickets
Semantic analytics	Certified metrics, dimensions, row-level security, query validation and lineage
Tool gateway	Typed tool registry, policy enforcement, idempotency, confirmation and approval hooks
Domain platform	Finance, sales, procurement, inventory, HR, MRP, service and shared services
Trust plane	Audit, telemetry, evaluations, redaction, retention, incident response and governance

4. AI Platform Components

Component	Specification
AI Gateway	Provider abstraction, model selection by task and risk, token budgets, caching, regional routing, fallback and circuit breaking.
Intent Router	Classifies help, navigation, analytics, draft, execution and automation intents; rejects unsupported or unsafe intents.
Agent Orchestrator	Coordinates specialist capabilities while presenting one Ask ERP identity to users. Plans are bounded by step, time and cost limits.
Context Engine	Builds a signed context envelope from current UI state, identity, permissions, company configuration and terminology.
Tool Registry	Publishes strongly typed, versioned tools with descriptions, schemas, permission requirements, risk level and idempotency policy.
Knowledge Service	Indexes official help, release-specific UI instructions, company SOPs and policies with tenant isolation and source citations.
Semantic Query Service	Translates questions only into approved measures and dimensions; it never gives the model unrestricted SQL.
Conversation Service	Stores threads, summaries, language preference, pending confirmations and tool results under explicit retention rules.
Evaluation Service	Runs golden tasks, multilingual tests, security probes, answer grounding checks and regression gates before releases.
AI Operations Console	Shows usage, latency, cost, errors, unsafe attempts, feedback, model versions, tool failures and rollback controls.

Specialized capabilities

Implement domain specialists—Sales, CRM, Procurement, Inventory, Finance, Accounting, HR, Payroll, Manufacturing, Service, Analytics, Approval and Help—as policies and tool bundles behind one orchestrator. Avoid independent chatbots with separate memory or inconsistent permissions.

Model strategy

- Small, fast model for classification, navigation and simple help.
- Stronger model for multi-step analysis, document interpretation and complex planning.
- Deterministic code for calculations, taxes, totals, posting, availability and permissions.
- Speech-to-text and text-to-speech selected per language and deployment region.
- No provider-specific contract leaks into domain modules; models are replaceable.

5. AI Safety, Action Levels & Trust

L1 READ Answer, explain, summarize No confirmation	L2 PREPARE Draft or stage a transaction User confirms	L3 EXECUTE Post, approve, pay, delete Strong confirmation + workflow
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Control	Required implementation
Authorization	Re-evaluate permissions at every tool call; never rely only on conversation context
Validation	Use the same domain validators as the conventional UI
Confirmation	Bind confirmation to an immutable action preview, user, context and expiration time
Approval	Create or advance the normal workflow; AI cannot self-approve
Idempotency	All mutation tools require an idempotency key and safe retry behavior
Audit	Log request, normalized intent, model/tool versions, arguments, result, confirmation and actor
Data minimization	Send only necessary fields; redact secrets and restricted personal data
Prompt defense	Treat documents, notes and imported text as untrusted data—not instructions
Separation of duties	Enforce maker-checker and financial controls regardless of channel
Kill switch	Disable model, tool, tenant or action class without deploying code

AI audit event

Minimum fields: tenant_id, user_id, session_id, conversation_id, source channel, locale, original request, normalized intent, context hash, model and prompt version, retrieved source IDs, planned tools, actual calls, arguments hash, result identifiers, risk level, confirmation evidence, workflow ID, latency, cost, outcome and error category.

Truth rules

- Never invent a customer, item, balance, posting status or approval result.
- A number without scope is incomplete.
- A forecast is labeled and includes assumptions.
- If data is stale or unavailable, say so and provide a refresh or drill-down action.
- Generated explanations do not override accounting records or legal documents.

6. Multilingual & Voice Blueprint

Localization is not limited to translated labels. The full interaction loop—speech, intent, business terminology, explanation, confirmation and generated output—must operate in the user's language.

Capability	Requirement
Language detection	Detect per message and allow explicit override
Code mixing	Understand combinations such as English + Malayalam without forcing language selection
Terminology memory	Map local terms such as party, godown and booking to canonical ERP entities
Business entities	Preserve names, SKUs, tax codes and legal text without unwanted translation
Voice input	Support noisy environments, confirmation playback and correction before action
Voice output	Read summaries and prompts; never speak sensitive values in insecure contexts
Numbers and dates	Confirm ambiguous dates, units, decimals and currency formats
Generated documents	Use approved multilingual templates and fonts
Evaluation	Maintain intent, entity and safety test sets for every supported launch language

Example interaction

User: Last month Kozhikode branch sales ethra?
AI: Last month Kozhikode branch net sales were Rs. 28.6 lakh, excluding cancelled invoices. That is 12.4% above the previous month.
Actions: View report | Compare branches | Export PDF | Explain change

Confirmation design

For monetary or inventory-impacting actions, repeat the key entity, quantity, warehouse, amount, date and destination in the user's language. Require an explicit affirmative response; ambiguous acknowledgements do not count.

7. AI in Every Business Module

Each module ships with four AI capability classes: Explain, Analyze, Prepare and Act. A module is not AI-ready until its tool contracts, semantic measures, help knowledge, permissions and evaluation cases are complete.

CRM & Customer

Core scope	AI-native capabilities	Controlled actions
Lead capture, customer 360, activities, pipeline, segmentation, credit profile and consent.	Summarize the relationship; rank follow-ups; detect churn; draft multilingual outreach; convert notes to activities; explain next best action.	Create lead/contact/task drafts; schedule follow-up; draft communication. Sending or changing credit requires permission/confirmation.

Sales

Core scope	AI-native capabilities	Controlled actions
Quotation, order, pricing, discount, availability, fulfillment, returns and invoicing.	Create quotations from conversation; explain price and margin; detect stalled orders; forecast pipeline; suggest cross-sell.	Draft quote/order/invoice. Enforce price lists, stock, tax, discount limits and approval thresholds.

Procurement

Core scope	AI-native capabilities	Controlled actions
Requests, RFQ, bid comparison, PO, receipt, bills and supplier performance.	Recommend purchasing; compare quotes; explain price variance; detect delay and concentration risk.	Draft PR/RFQ/PO. Supplier selection and PO release follow approval and budget rules.

Inventory & WMS

Core scope	AI-native capabilities	Controlled actions
Items, units, warehouses, bins, batches, serials, reservations, transfers, counts and valuation.	Explain stock; predict stock-out; flag slow/dead/expiry stock; recommend replenishment and put-away.	Prepare transfers, counts and adjustments. Posting requires location, quantity and batch validation.

Finance & Accounting

Core scope	AI-native capabilities	Controlled actions
GL, AR, AP, tax, banking, reconciliation, budgets, assets and close.	Explain profit/cash/variance; prioritize collections; match bank items; find anomalies; assist period close.	Draft journals, receipts and payments. Posting/payment are L3 with maker-checker and immutable audit.

HR & Payroll

Core scope	AI-native capabilities	Controlled actions
Employees, recruitment, attendance, leave, expenses, payroll and performance.	Answer policy questions; explain leave/pay; identify overtime and attendance exceptions; summarize candidates.	Prepare leave, expense and payroll changes. Sensitive access and final payroll require strict roles.

Manufacturing & Quality

Core scope	AI-native capabilities	Controlled actions
BOM, routing, MRP, plans, work orders, consumption, output, quality and maintenance.	Feasibility checks; shortage and capacity explanations; scrap anomaly detection; maintenance prediction.	Draft plans/WO/material requests. Release and completion follow capacity, stock and quality gates.

Projects & Timesheets

Core scope	AI-native capabilities	Controlled actions
Projects, tasks, resources, time, cost, billing, milestones and profitability.	Summarize status; predict delay/budget risk; draft plans; explain profitability.	Prepare tasks, assignments, timesheets and billing drafts within project controls.

Service & Field Operations

Core scope	AI-native capabilities	Controlled actions
Tickets, SLA, warranty, AMC, dispatch, visit, parts and resolution.	Summarize history; classify issue; suggest resolution; detect SLA breach; optimize technician assignment.	Draft response, visit and parts request. Customer communication and closure require review rules.

POS & Commerce

Core scope	AI-native capabilities	Controlled actions
Catalog, checkout, payments, returns, loyalty, promotions, channels and reconciliation.	Explain promotion; recommend products; flag suspicious returns; summarize channel performance.	Prepare returns or overrides; cashier policy and supervisor approval remain authoritative.

Logistics & Fleet

Core scope	AI-native capabilities	Controlled actions
Delivery, shipment, route, proof, vehicle, driver, fuel and maintenance.	Predict delay; optimize route/load; explain delivery exceptions and fleet cost.	Prepare dispatch/route updates. High-impact rerouting and expense posting require confirmation.

Contracts & Documents

Core scope	AI-native capabilities	Controlled actions
Templates, versioning, metadata, renewal, obligations, e-sign and archive.	Extract terms; summarize obligations; compare versions; alert renewal and compliance risk.	Draft documents and tasks. Legal approval and signatures remain human-governed.

8. Owner, Admin & Manager Intelligence

The executive experience turns certified ERP metrics into an interactive management conversation. It must answer what happened, why, what may happen next and what action is available.

Daily business brief

Section	Default measures	AI contribution
Trading	Sales, returns, gross margin, orders	Compare trend and explain material movement
Cash	Receipts, payments, bank, forecast	Highlight liquidity risk and upcoming obligations
Working capital	AR/AP aging, inventory days	Prioritize collections, payments and excess stock
Operations	Open deliveries, shortages, delays	Identify bottlenecks and responsible owners
People	Attendance, leave, overtime	Flag exceptions within privacy policy
Decisions	Pending approvals and breaches	Produce concise decision packs with risk/context

Conversational report sequence

- Show customers who purchased above Rs. 1 lakh in the last 90 days but not this month.
- Only Kerala; sort by sales; add outstanding balance.
- Explain the top five overdue cases.
- Save this as a report and send it to the Sales Manager every Monday.

Admin copilot

- Configure modules and menus from a guided business interview.
- Generate draft roles and permissions from job descriptions; admin reviews differences.
- Convert natural-language approval rules into a visual workflow preview.
- Create dashboards from requested responsibilities and certified KPIs.
- Build trigger-condition-action automations with collision and recursion checks.
- Explain configuration impact before activation and preserve version history.

Decision pack contract

Every approval summary includes request, amount, budget effect, policy status, prior comparable transactions, alternatives, exceptions, risk factors, source links and available actions. AI recommendations are advisory and never conceal required approvers.

9. Reporting & Semantic Data Layer

Conversational analytics must sit above a governed semantic layer. The model selects certified measures and dimensions; the query service applies row-level access, validates combinations and returns a structured result with lineage.

Domain	Certified measures	Common dimensions
Sales	Gross/net sales, returns, margin, order value	Date, customer, item, salesperson, branch, channel
Finance	Revenue, COGS, expense, profit, cash, AR/AP	Period, account, cost center, project, currency
Inventory	On hand, available, reserved, value, turns	Item, category, warehouse, bin, batch, age
Procurement	Spend, variance, lead time, fill rate	Supplier, item, buyer, branch, period
HR	Headcount, absence, overtime, payroll cost	Department, role, location, period
Manufacturing	Output, yield, scrap, utilization, variance	Plant, work center, product, order, shift

Query safeguards

- Allowlisted measures, joins, filters and date grains only.
- Automatic tenant and row-level filters cannot be overridden by the model.
- Query complexity, result size and export limits are enforced.
- Numeric calculation is deterministic and reproducible.
- Generated narrative is separated from the returned data table.
- Saved reports store semantic definitions—not model-generated raw SQL.

Answer anatomy

- Headline answer with units and currency.
- Scope chip: company, branch, period, status and comparison basis.
- Small supporting table or chart when useful.
- Explanation of major drivers, labeled as calculated or inferred.
- Drill-down link to the governed report.
- Actions: change filter, export, schedule or create follow-up task.

10. Core Platform & Conventional ERP Foundation

Shared service	Required scope
Organization	Companies, branches, divisions, departments, locations, cost/profit centers, fiscal periods, currencies, tax and numbering.
Identity & access	SSO/MFA, users, roles, ABAC/RBAC, sessions, device controls, field/record access and separation of duties.
Workflow	Conditions, thresholds, multi-level approval, delegation, escalation, resubmission, versioning and history.
Notifications	In-app, email, SMS, WhatsApp and push; preferences, templates, quiet hours, delivery status and opt-out.
Master data	Customers, suppliers, items, employees, accounts, tax, warehouses, duplicates, approvals, effective dating and stewardship.
Documents	Attachments, templates, OCR, classification, versioning, retention, e-signature and virus scanning.
Search	Global entity search, permission-aware indexing, saved searches and recent items.
Import/export	Mapping, validation, dry run, error file, resumable batches, audit and controlled exports.
Integration	Versioned REST/event APIs, webhooks, connector framework, retries, dead-letter queue and observability.
Audit	Append-only security, configuration, transaction and AI events with before/after values.

Accounting integrity

Operational modules post through defined accounting rules to balanced journal entries. Posted records are reversed, not silently edited. Period locks, tax determination, multi-currency valuation, numbering, inventory valuation and consolidation remain deterministic services.

Multi-company tenancy

Tenant isolation is mandatory. A user may operate across permitted companies, but context is explicit and every query/action is scoped. Shared masters require controlled ownership, replication and conflict policy.

11. Data Architecture & Key Entities

Domain	Representative entities
Identity	Tenant, User, Role, Permission, Policy, Session, Device, Consent
Organization	Company, Branch, Department, CostCenter, FiscalPeriod, Currency, ExchangeRate
Commercial	Lead, Contact, Customer, Opportunity, Quote, SalesOrder, Invoice, Receipt, Return
Supply	Supplier, PurchaseRequest, RFQ, Bid, PurchaseOrder, GRN, SupplierBill, Payment
Inventory	Item, UOM, Warehouse, Bin, Lot, Serial, StockLedger, Reservation, Transfer, Count
Finance	Account, Journal, JournalLine, TaxCode, Budget, BankTransaction, Reconciliation, Asset
People	Employee, Contract, Shift, Attendance, Leave, Expense, PayrollRun, Payslip
Production	BOM, Routing, Plan, WorkOrder, MaterialIssue, Output, QualityCheck, MaintenanceOrder
Service	Ticket, SLA, Entitlement, Visit, Technician, Resolution, Warranty
AI	Conversation, Message, ContextEnvelope, ToolDefinition, ToolCall, Confirmation, Evaluation, Feedback

Data rules

- Every business table carries tenant_id; high-volume tables also carry company and partition keys.
- Use immutable ledger patterns for stock and accounting; balances are derived or maintained with reconciliation controls.
- Effective-dated configuration supports tax, pricing, organization and policies.
- Personally identifiable and financial data are classified, encrypted and retained by policy.
- Vector indexes contain tenant-scoped representations and source metadata; deletion propagates.
- AI conversation data is not treated as the system of record.

Event model

Publish versioned events such as SalesOrderConfirmed, StockReserved, GoodsReceived, InvoicePosted, PaymentAllocated, ApprovalRequested and PayrollCompleted. Consumers use idempotent processing, correlation IDs, outbox pattern and dead-letter handling.

12. Tool & API Contract Blueprint

Tools are thin, typed contracts over domain application services. Each tool declares risk, permissions, required context, validation, confirmation policy, idempotency and observable outcomes.

Field	Example / rule
name	sales.create_quotation_draft.v1
purpose	Create a non-posting quotation draft
input	customer_id, items[], requested_discount, validity_date, currency
context	tenant_id, company_id, branch_id, user_id, locale
permission	sales.quotation.create
risk	L2 Prepare
validation	Customer active; items sellable; price/tax available; credit warning returned
output	draft_id, totals, warnings, approval_requirement, preview_version
confirmation	Required before submission; confirmation binds preview_version
idempotency	Required for create/submit tools
audit	Original intent, resolved IDs, input/output hashes and result reference

Tool design rules

- Prefer business-level tools over CRUD primitives.
- Return structured warnings and next valid actions—not only error strings.
- Separate draft, validate, submit, approve, post and reverse operations.
- Never allow arbitrary table names, field names, code execution or unrestricted queries.
- Keep tool outputs compact; retrieve detailed records only when needed.
- Version schemas and retain backward compatibility during model rollout.

Example catalog

Read	Prepare	Execute
get_customer_balance	create_quotation_draft	submit_quotation
check_stock_availability	prepare_stock_transfer	post_stock_transfer
sales_summary	prepare_sales_report	schedule_report
get_leave_balance	prepare_leave_request	submit_leave_request
explain_journal	prepare_journal_draft	post_journal (restricted)

13. Security, Privacy & Compliance

Threat / concern	Controls
Cross-tenant leakage	Tenant-bound identity, scoped retrieval, policy enforcement, test probes and isolation monitoring
Prompt injection	Content sanitization, instruction/data separation, tool allowlists and no secret exposure
Excessive agency	Step/time/tool limits, action levels, confirmation, approvals and kill switches
Unauthorized reporting	Semantic row/field security, export controls and watermarking
Sensitive data exposure	Classification, masking, DLP, encrypted transport/storage and least-data prompts
Fraud / SoD bypass	Maker-checker, transaction limits, device/risk checks and independent approval
Model/provider risk	No training by default, contractual controls, regional endpoints, retention settings and provider fallback
Untraceable output	Source IDs, prompt/tool/model versions, decision logs and reproducible numeric queries
Account compromise	MFA, SSO, session/device controls, anomaly detection and step-up authentication
Availability	Circuit breakers, degraded conventional UI, queues, recovery targets and provider failover

Compliance posture

Design for applicable privacy, tax, accounting, employment and electronic-record requirements in each deployment country. Configuration and legal review determine actual obligations. The AI layer must not present legal or accounting interpretation as authoritative policy.

Retention

Define separate retention for business records, audit logs, AI conversations, model traces, retrieved documents and telemetry. Users receive visibility appropriate to their role; administrators can apply legal hold and deletion policies without breaking statutory records.

14. Non-Functional Requirements

Area	Baseline target
Availability	Core transactional ERP 99.9% monthly; AI may degrade independently without blocking conventional work
Performance	P95 normal UI reads <2s; writes <3s excluding external integrations; AI streaming begins <2s where feasible
Scale	Horizontal services; partition high-volume ledgers/events; tenant quotas and workload isolation
Consistency	Strong consistency for posting, stock and approvals; eventual consistency for search, BI and notifications
Recovery	Defined RPO/RTO by service tier; tested backups, point-in-time recovery and regional disaster procedure
Observability	Metrics, logs, traces, correlation IDs, business events, AI token/cost/tool telemetry and synthetic tests
Accessibility	Keyboard navigation, semantic labels, contrast, text scaling and screen-reader-compatible guided help
Mobile/offline	Responsive priority flows; queued field operations with conflict handling where approved
Maintainability	Modular monolith initially where practical; explicit domain boundaries and versioned contracts
Localization	Unicode, RTL readiness, timezone, fiscal/calendar, currency, units, numbering and document templates

Degraded mode

If AI is unavailable, users retain navigation, forms, reports and approvals. Pending AI drafts remain identifiable and recoverable. The system explains the limitation without pretending an action succeeded.

15. Delivery Architecture & Environments

Concern	Recommendation
Application shape	Start with well-modularized domain services or modular monolith; isolate ledgers, identity, AI gateway and integration workloads as needed
Frontend	Component system with schema-driven forms, action metadata, guided overlays and consistent Ask ERP panel
Backend	Domain application services, workflow engine, event outbox, background jobs and integration gateway
Data	Relational system of record; cache; object storage; search index; warehouse/lakehouse for analytics; vector index for knowledge
Deployment	Containerized workloads, infrastructure as code, secrets manager, automated migrations and rolling/canary releases
Environments	Local, dev, test, staging, UAT, production; isolated data and model credentials
Release	Feature flags by tenant/module/tool/model; progressive AI rollout and instant rollback
Testing	Unit, contract, integration, ledger invariants, workflow, migration, performance, security and AI evaluations

Build vs buy

Buy commodity capabilities such as identity federation, communications and model APIs where appropriate. Own the ERP domain model, tool contracts, semantic metrics, permissions, audit, workflows, terminology and evaluation assets—the parts that create product differentiation and trust.

16. Implementation Roadmap

Phase	Indicative duration	Exit scope
Phase 0 - Discovery & controls	4-6 weeks	Personas, top workflows, target industries, legal/localization scope, data classification, architecture decisions, success metrics and AI risk policy.
Phase 1 - Platform + AI foundation	10-14 weeks	Organization, identity, permissions, workflow, audit, master data, event backbone, design system, Ask ERP shell, context engine, tool gateway, multilingual help and evaluation harness.
Phase 2 - Revenue & stock MVP	14-18 weeks	CRM, sales, procurement, inventory, essential finance postings, certified reports; AI help/read/drafts for the top workflows; pilot data migration.
Phase 3 - Finance depth & operations	12-16 weeks	GL/AP/AR/tax/banking/budget/assets, reconciliation, close assistant, collections intelligence, mobile warehouse and approvals.
Phase 4 - People, projects & service	12-16 weeks	HR, attendance, leave, expenses, payroll, projects, helpdesk and field service with role-specific copilots.
Phase 5 - Manufacturing & commerce	16-24 weeks	BOM/MRP/work orders/quality/maintenance, POS, e-commerce, logistics and advanced planning intelligence.
Phase 6 - Scale & ecosystem	Continuous	Industry packs, partner APIs, marketplace, advanced forecasting, agentic automation, country localizations and enterprise compliance.

Release gate for every module

- Conventional UI workflow complete and tested.
- Permissions, validations, workflow, audit and accounting integration complete.
- Help knowledge and contextual guidance complete.
- Certified metrics and drill-downs complete.
- Read and draft tools implemented with contracts and idempotency.
- Golden task, multilingual, security and regression evaluations pass.
- Telemetry, cost budget, fallback and support playbooks are active.

Pilot strategy

Pilot with 2-3 businesses in one or two closely related industries. Start with a limited language set and the top 25 tasks. Capture task completion, corrections, unsupported intents, latency and trust feedback weekly; expand capability only after the underlying workflows are stable.

17. Team, Governance & Operating Model

Workstream	Core ownership
Product	ERP product lead, domain product managers, UX research and rollout
Domain engineering	Sales/procurement/inventory, finance, people/service and manufacturing teams
AI platform	Gateway, orchestration, tools, knowledge, semantic analytics and evaluations
Experience	Design system, web/mobile, accessibility, guided UI, conversation and voice
Data	Operational modeling, warehouse, metrics, governance, quality and lineage
Trust	Security, privacy, compliance, fraud controls, red team and incident response
Implementation	Migration, configuration, integrations, training, partner/customer success and support
Quality	Automation, performance, accounting invariants, localization and AI regression

AI governance board

- Approves action classes, restricted tools and model/provider changes.
- Reviews safety incidents, hallucination patterns, customer feedback and cost trends.
- Owns release thresholds for multilingual and domain evaluations.
- Maintains the risk register, kill-switch procedure and exception process.
- Ensures legal, accounting and HR experts review sensitive capabilities.

RACI rule

AI may recommend; an accountable product or domain owner approves the capability. Business users remain accountable for confirmed transactions, and approvers remain accountable for approvals. Operational incidents have named on-call ownership.

18. Testing & Evaluation Blueprint

Test family	Examples	Release threshold
Functional	Tool schemas, validations, workflows, ledger postings	All critical paths pass
Golden conversations	Known intents, follow-ups, ambiguity and action previews	Task success target met
Grounding	Help citations, metric scope, source drill-down	No unsupported material claims
Multilingual	Intent/entities/code mixing/numbers/voice	Per-language target met
Authorization	Role, branch, company, field and record restrictions	Zero prohibited access
Adversarial	Prompt injection, exfiltration, tool abuse, social engineering	No critical escape
Accounting	Balanced journals, tax, valuation, closing and reversal invariants	Zero invariant violations
Reliability	Retries, timeout, provider failure, duplicate messages	No duplicate side effects
Performance	Concurrent UI, analytics, tool and model workloads	P95 targets met
Human factors	First-time user completion and correction behavior	Top workflows meet usability goal

Evaluation dataset

Build versioned test suites from synthetic scenarios, approved anonymized pilot cases, domain edge cases and production failures. Each case specifies persona, permissions, context, language, expected tools, prohibited behavior, acceptable answer properties and final state.

Production feedback

Collect explicit helpful/not-helpful feedback and operational signals such as correction, abandonment, undo, escalation and tool failure. Do not equate user acceptance alone with correctness; financial and security outcomes require independent checks.

19. Initial Product Backlog

Priority	Epic	Definition
P0	Ask ERP shell	Global command bar, screen panel, streaming, language preference and conversation history
P0	Context envelope	Identity, role, tenant, screen, record, permissions and signed context version
P0	Tool gateway	Registry, schema validation, authz, risk, confirmation, idempotency and audit
P0	Help copilot	Release-specific knowledge, citations, Teach Me overlays and error explanations
P0	Semantic analytics	Certified sales, finance and inventory measures with row security and drill-down
P0	Core domains	Organization, users, workflow, audit, CRM, sales, procurement, inventory and finance postings
P0	Draft actions	Customer, lead, quotation, sales order, purchase request, transfer and leave request
P0	AI operations	Usage, latency, failure, evaluation, feedback, cost and kill switches
P1	Owner brief	Daily business summary, exceptions, approvals and proactive insights
P1	Voice	Speech input/output for selected high-value mobile workflows
P1	Admin copilot	Guided setup, workflow generation, dashboards and automation previews
P1	Report builder	Conversational filter refinement, save, export and scheduling
P1	Collections	Prioritization, reminder drafts and collection task creation
P2	Forecasting	Cash, demand, stock-out and capacity predictions with backtesting
P2	Industry packs	Terminology, workflows, masters, KPIs and AI evaluation packs

Recommended first 25 workflows

Customer creation; lead follow-up; quotation; sales order; invoice; receipt; customer balance; stock lookup; transfer; adjustment draft; low-stock review; purchase request; RFQ comparison; purchase order; goods receipt; supplier bill; payment draft; journal explanation; bank matching; sales report; overdue report; leave balance/request; expense claim; approval decision pack; contextual how-to guidance.

20. Acceptance Criteria for the AI-Native Promise

Promise	Acceptance criteria
Easy usage	A first-time user can complete the top workflows through guided interaction with no external manual
AI on every screen	The assistant knows module/screen/record context and offers relevant help, analysis and safe actions
Mother tongue	Configured languages support help, reporting and confirmation while preserving exact business entities and numbers
Admin asks for reports	Natural language produces governed, reproducible results with scope, lineage, drill-down and export
AI performs work	Supported actions create validated drafts or governed transactions through typed domain tools
No rule bypass	Permissions, validation, workflow, separation of duties and audit match the conventional UI
Trustworthy numbers	Every important metric is certified, scoped and reconcilable to the system of record
Explainable failure	When a request cannot be completed, the user receives the exact blocker and a safe next step
Operationally manageable	Admins can monitor, restrict, evaluate, roll back and disable AI capabilities by tenant/tool/action
Resilient ERP	Core business operations continue if model providers or AI services are unavailable

Final architecture decision

Build the secure AI interaction and action layer in Phase 1. Every later ERP module must publish governed tools, certified metrics, contextual help and evaluation cases as part of its definition of done. This prevents the product from becoming forty conventional modules with a decorative chatbot.

Product statement

Ask. Understand. Act. — in the language of the user, under the controls of the business.

Appendix A. Sample AI Action Flow

Request: 'Create a quotation for Rahman Traders: 50 boxes Product A and 20 boxes Product B. Give 3% discount.'

Step	System behavior
1. Understand	Classify create-quotation intent; extract customer, items, quantities and discount
2. Resolve	Find permitted customer/item candidates; ask only if ambiguous
3. Authorize	Check quotation-create access for the active company and branch
4. Enrich	Load price list, UOM, tax, currency, stock, customer credit and discount policy
5. Validate	Calculate totals deterministically; identify that 3% exceeds the user's 2% limit
6. Preview	Show exact lines, taxes, availability, margin warning and required manager approval
7. Confirm	User confirms the bound preview
8. Execute	Create quotation draft and submit the discount exception to normal workflow
9. Audit	Record request, resolution, tool calls, preview, confirmation and result IDs
10. Respond	Return draft number, approval status and actions: view, edit, withdraw or follow approval

Failure behavior

If a product is ambiguous, stock is unavailable or tax cannot be determined, the system stops before creation, explains the specific issue and offers constrained choices. It never guesses a legal or monetary field.

Appendix B. Example AI Audit Record

Actor: user_ahmed | Tenant: tenant_018 | Company: company_kozhikode
Request: Create quotation for ABC customer
Intent: sales.create_quotation | Risk: L2 Prepare
Context: Sales / Quotation / new | Locale: ml-IN
Tool: sales.create_quotation_draft.v1
Arguments: customer_id=382; items=[...]; discount=2%; currency=INR
Validation: PASS with stock warning | Preview version: pv_7392
Confirmed: YES | Confirmation time: 2026-09-22 14:43:21 UTC
Result: QT-2026-00192 | Outcome: Draft created
Model / prompt / tool versions: retained | Correlation ID: corr_6f8...

Required properties

- Append-only and protected from ordinary application edits.
- Searchable by user, business result, conversation, tool, date and correlation ID.
- Sensitive fields masked according to viewer permission.
- Linked to conventional transaction/workflow audit history.
- Retention and export comply with the tenant's legal and security policy.